



Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: B.E., V & ECE A

Course Code & Title: EC8551 & Communication Networks

Name of the Faculty member (s): Ms.S.Harini Shriram

Innovative Practice Description

- **Unit / Topic:** V/ Electronic Mail
- **Course Outcome:** CO : Students able to classify different kind of application layer protocols
- **Topic Learning Outcome:** TLO: Explain the concept of mail transfer protocol: SMTP, POP, IMAP and MIME.

- **Activity Chosen:** Flipped class room

- **Justification:** It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning.

- **Time Allotted for the Activity:** 40 minutes

- **Details of the Implementation:**

Specific topic was given to the students learn on their own. Resources like reference book, videos were given to the students. Students are asked to prepare more in depth than before. Students are separate into groups where students are given a task to perform. Get the class back together to share the individual group's work with everyone.

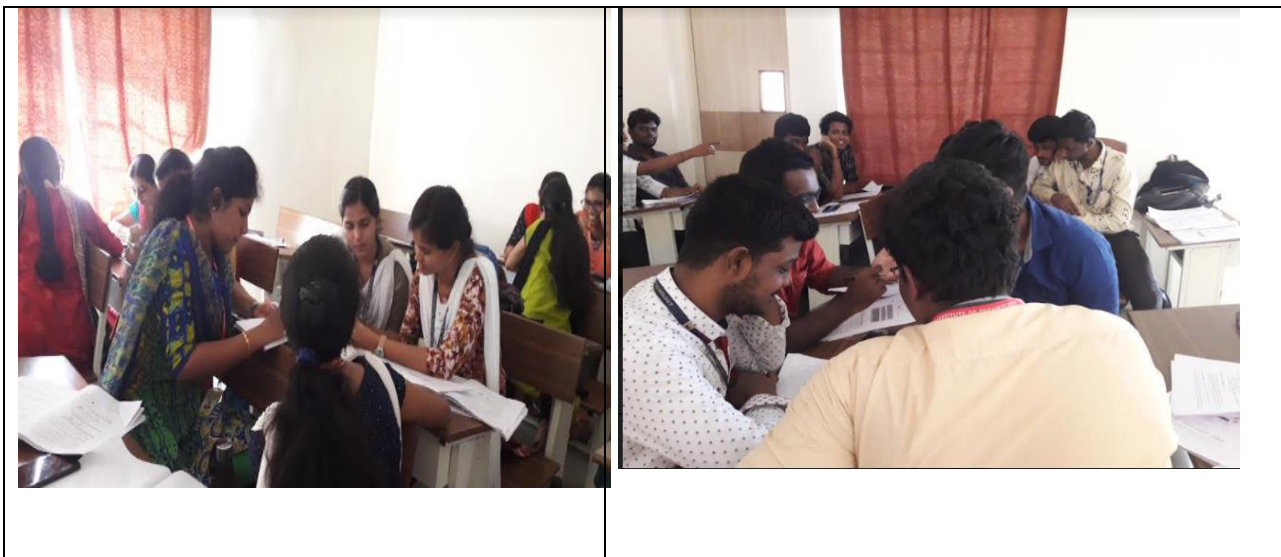
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO12	PSO1
CO5	2	3	2	1	1	1	1	1	1	2

- **PO / PSO mapped:**

Innovative practice	PO8	PO9	PSO1
	1	1	2
Justification for correlation	Identify tenets of the professional code of ethics while doing activity-based learning. So, course outcome is mapped with level 1	As team students prepares for flipped class room activity, it enables students to learn and diverse the content as an individual and team; hence course outcome is mapped with level 1.	Students will be able to adapt with new development electronics mail protocols, so it is mapped with level 2.

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Self-learning and accountability of the students got improved. They learn how to communicate with team members and work together to achieve common goals.

- ❖ ***Benefit of the practice:***

Every student got equal opportunity to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is that few students not exposed to flipped class room. Students struggle with self-discipline and may turn up to class without having absorbed the lesson. Few students hesitated to come forward and to interact. I have encouraged them to come forward and share their points

References:

- ❖ https://www.rcboe.org/cms/lib/GA01903614/Centricity/Domain/15451/Flip_your_Classroom.pdf
- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.
- ❖ Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from <https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html>.



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Innovative Practice Description

- **Unit / Topic:** IV /Transmission Control Protocols
- **Course Outcome:** CO : The students will be able to classify different kind of transport layer protocols
- **Topic Learning Outcome:** TLO : Explain the concept of Transmission Control Protocol and its services.
- **Activity Chosen:** Jigsaw
- **Justification:**
 - The Jigsaw technique is a cooperative learning method that bring about both individual accountability and achievement of the team goals. In this activity students depend on each other to succeed. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal;
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**

The topic is divided into subcategories. Students are divided into groups of five or six students. Then each group would be formed with one student receiving one subcategory of the topic. For this method, each group gets the same set of subcategories. Once individuals have researched and referred their subcategory, they will get together with individuals from the other groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.

CO – PO / PSO mapping:

CO	PO1	PO2	PO4	PO8	PO9	PO10	PSO1
CO4	2	2	1	1	1	1	2

PO / PSO mapped:

Innovative practice	PO1	PO2	PO4
	2	2	1
Justification for correlation	Applying communication strategies to identifying the	• Identification of engineering concepts such as data communication and handshaking technique	• The course outcome is mapped at level 1, because students will be able to understand the transport layer



	appropriate end to end protocol for reliable communication; thus course outcome is mapped with level 2.	to ensure reliable communication between networks; hence course outcome is mapped with level 2.	protocols and choose appropriate protocol for user specific applications.
Innovative practice	PO8	PO9	PSO1
	1	1	2
Justification for correlation	Identify tenets of the professional code of ethics while doing activity based learning. So, it is mapped at level 1	It helps the students to present results as a team, with smooth integration of contributions from all individual efforts. So, it is mapped at level 1	The student will be able to analyse and create flow control and error control techniques in communication networks; hence course outcome is mapped with level 2

• **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students gave positive feedback about the activity. They said it was interesting and knowledge improved on the topic. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 85% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

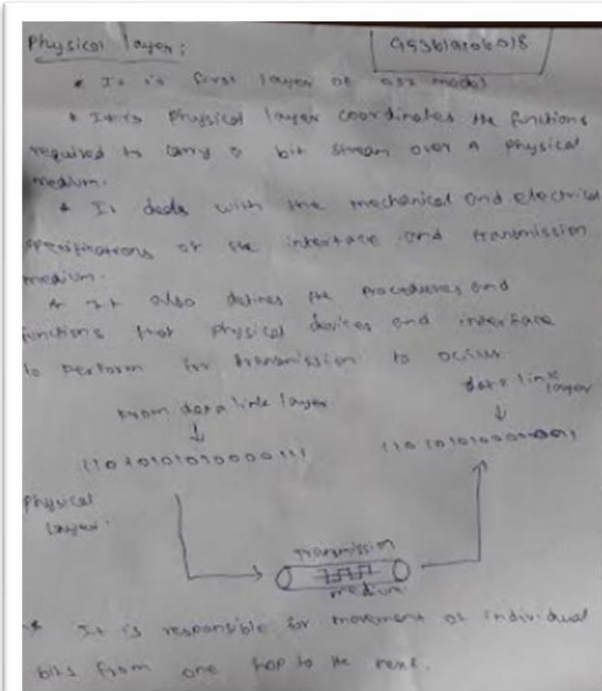
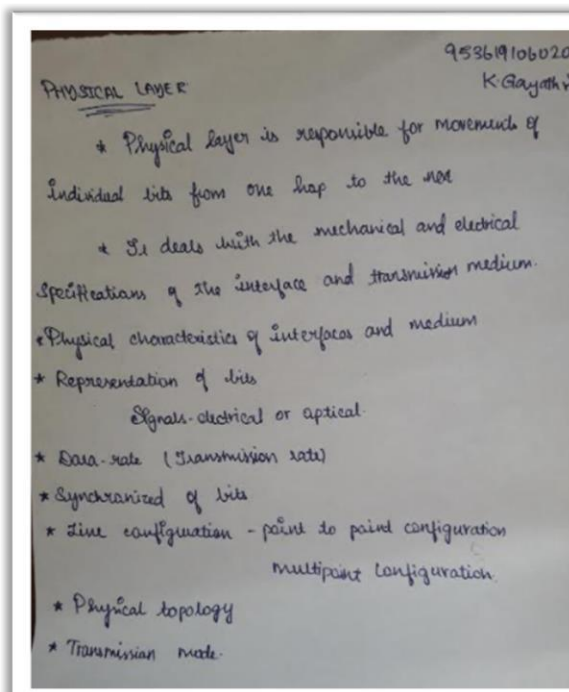
References:

- ❖ <https://www.readingrockets.org/strategies/jigsaw>
 - ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2021-2022 (ODD Semester)

INNOVATIVE TEACHING METHOD PLAN

Degree, Semester & Branch: V Semester B.E.ECE A
Course Code & Title: EC8551 & Communication Networks
Name of the Faculty member: Ms.S.Harini Shriram

UNIT I			
1.	25.08.2021	OSI Model-Physical Layer	Minute paper
			
2	27.08.2021	Overview of Data and Signals	Visible quiz
			

3	6.9.2021	Error Detection and Correction	Zero Minute Speech
			

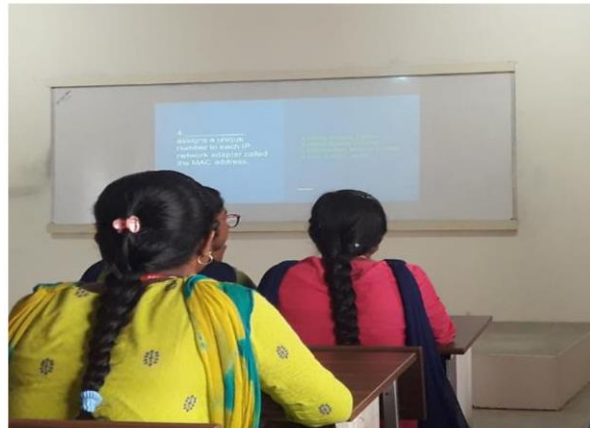
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UNIT II			
1.	16.09.2021	Wireless LANs- Available Protocols	Class Poll
			
2	20.9.2021	Zigbee	Think –Pair -Share
			

3	25.9.2021	Network Layer protocols – IP, ICMP	Story Board
			

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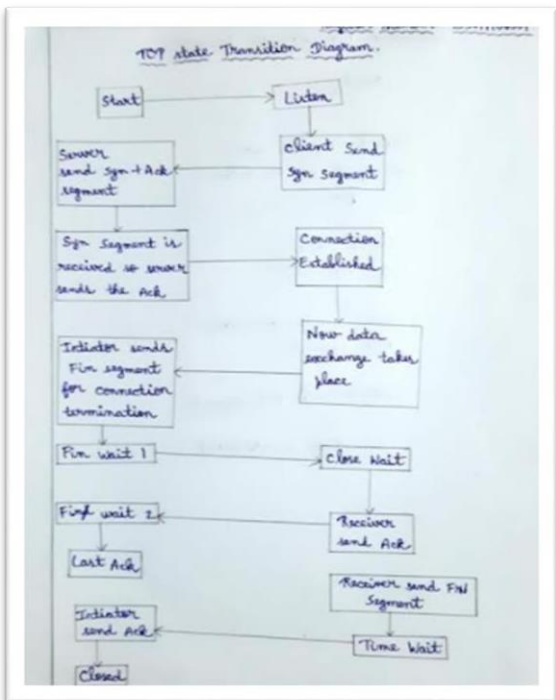
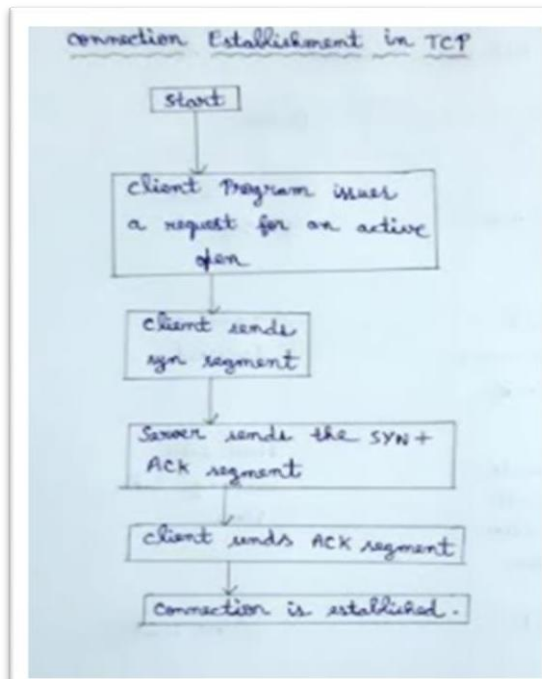
UNIT III			
1.	1.10.2021	Introduction to Routing	Informal Group
			
2	11.10.2021	Overview of Intra domain	Round Table
			

3	18.10.2021	Overview of IPv6 Addressing	Concepts Tests
			

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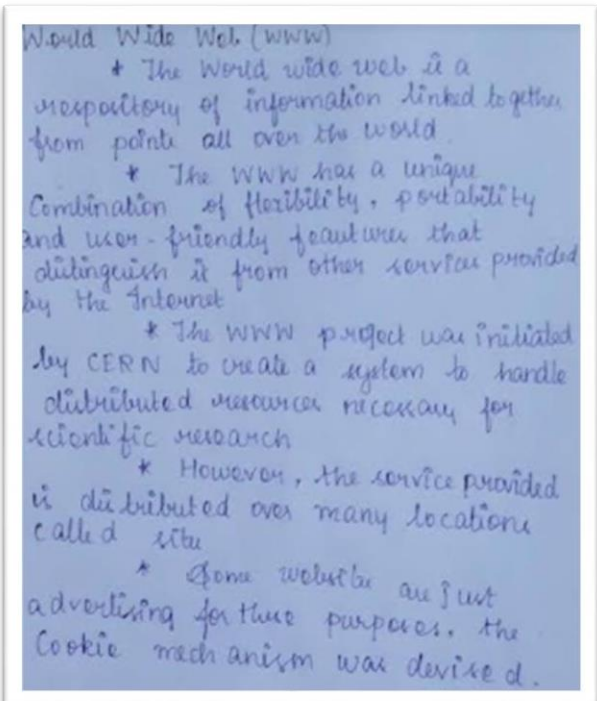
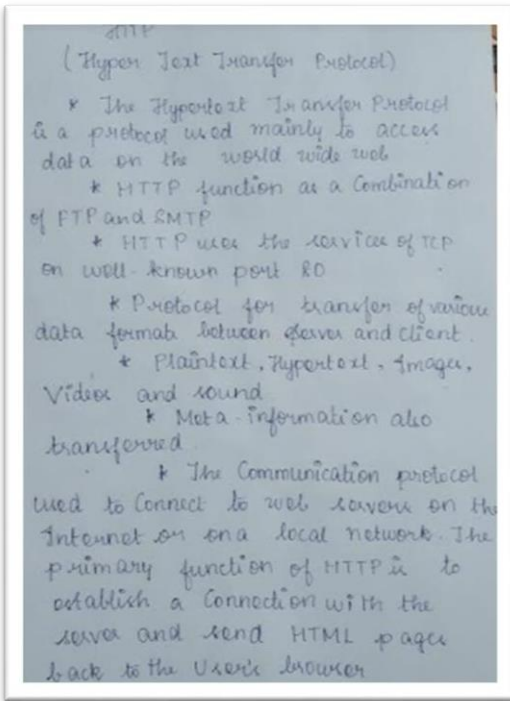
UNIT IV			
1.	25.10.2021	Transmission Control Protocol (TCP)	Informal Group Discuss
			
2	1.11.2021	TCP Connection - State transition diagram	Mind Map
			

3	18.11.2021	Application requirements	Library& Open Book Test
			

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UNITY			
1.	20.11.2021	World Wide Web, HTTP	Exit Slip
			
2	21.11.2021	DNS	Finger Signal
			

3	23.11.2021	Firewalls	Animated Video
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